Fraud Detection on Tabular Data Ensemble Pipeline – team 4

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Problem Setting

- **Input:** customer/application-related tabular features
- **Output:** fraud probability
- **Task type: binary classification**
- **Key challenge:**

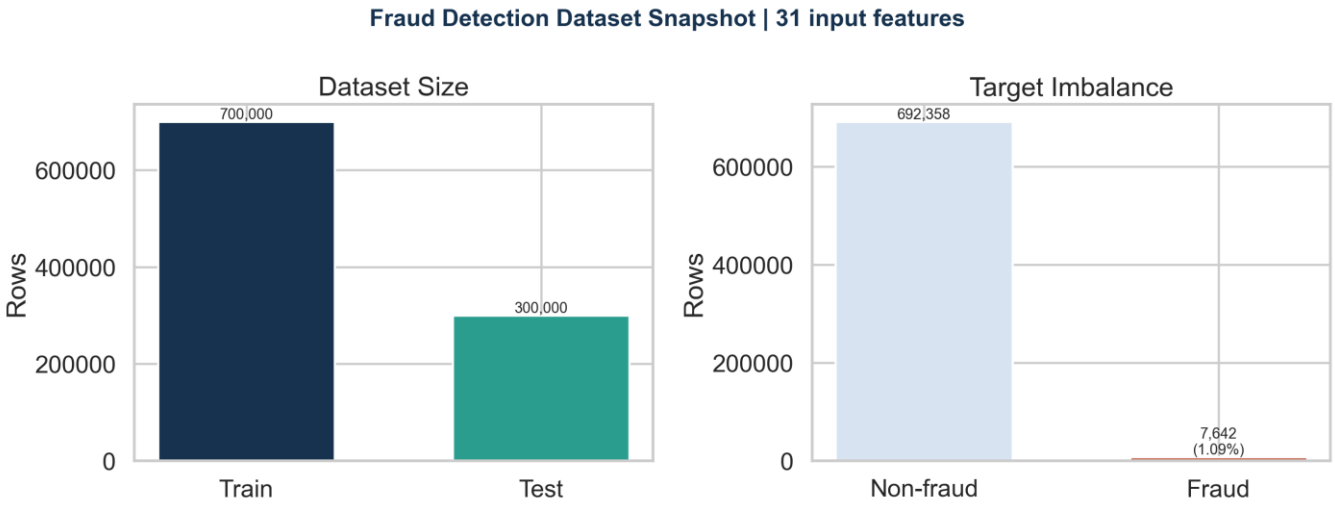
  extreme class imbalance

  numeric + categorical 혼합 구조

  일부 결측이 -1 placeholder로 표현됨

Accuracy is not enough; ranking fraudulent cases is the real objective.

Dataset Snapshot

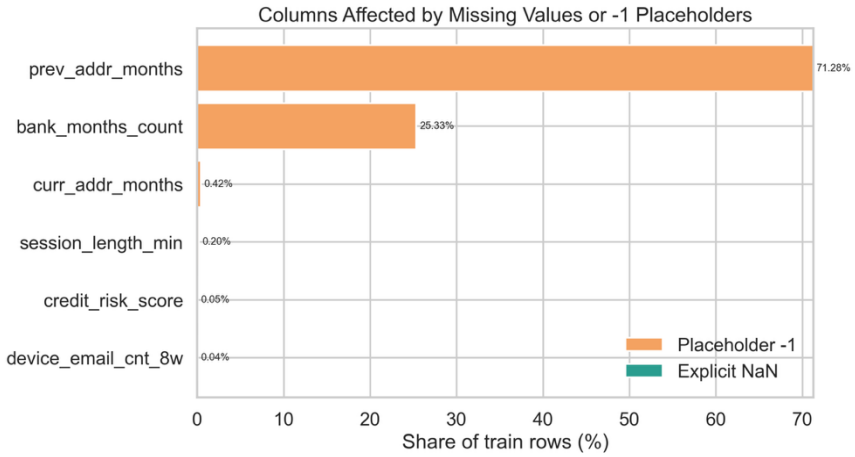


Train: **700,000**
Test: **300,000**
Fraud ratio: **1.09%**
매우 강한 **imbalanced classification** 문제
Main evaluation metric: **Average Precision / PRAUC**

The dataset size is large, but the positive class is extremely rare.

Missing Placeholder Analysis

Why -1 Was Treated as Missing



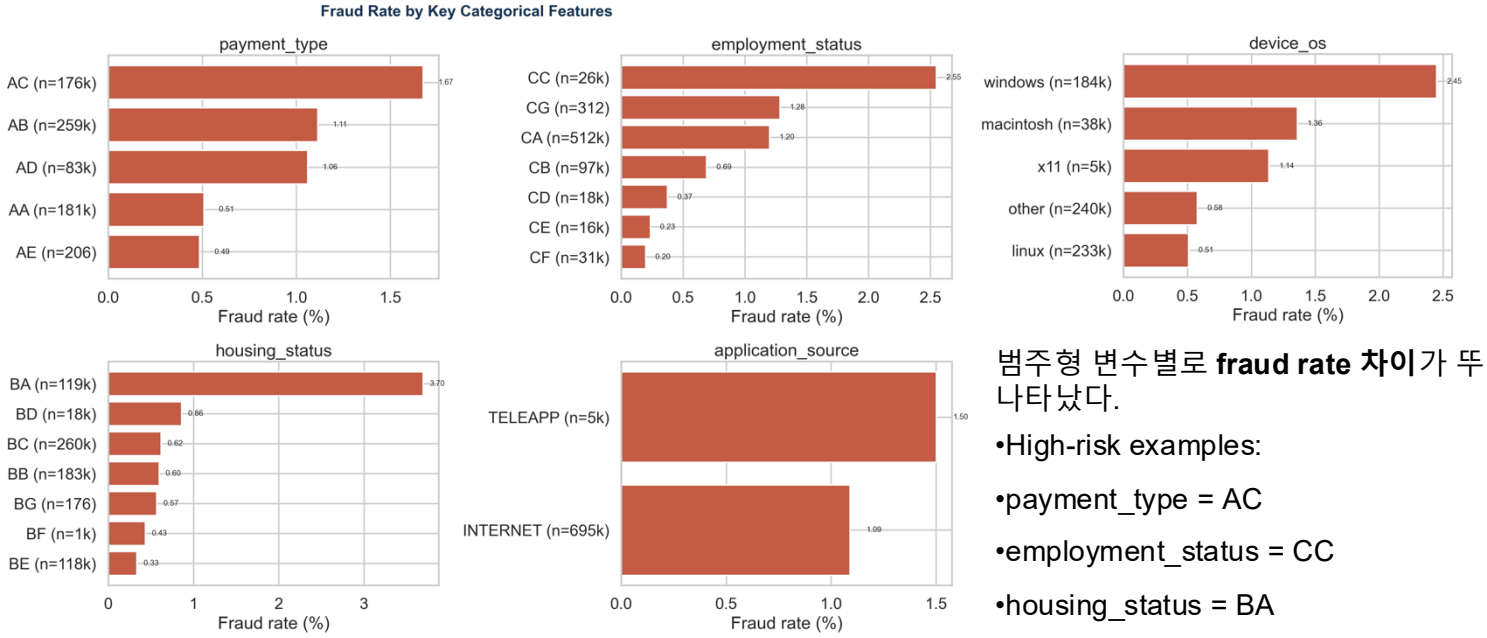
일부 column에서 -1은 자연스러운 수치값이 아니라 **missing placeholder**에 가깝다.

- Major examples:
- prev_addr_months: **71.28%**
- bank_months_count: **25.33%**
- 따라서 -1을 실제 숫자가 아니라 **정보 없음(unknown)**
- Converted to NaN during preprocessing

In these columns, -1 behaves more like “unknown” than an actual number.

Categorical Risk Profile

Categorical Variables Were Strong Fraud Signals



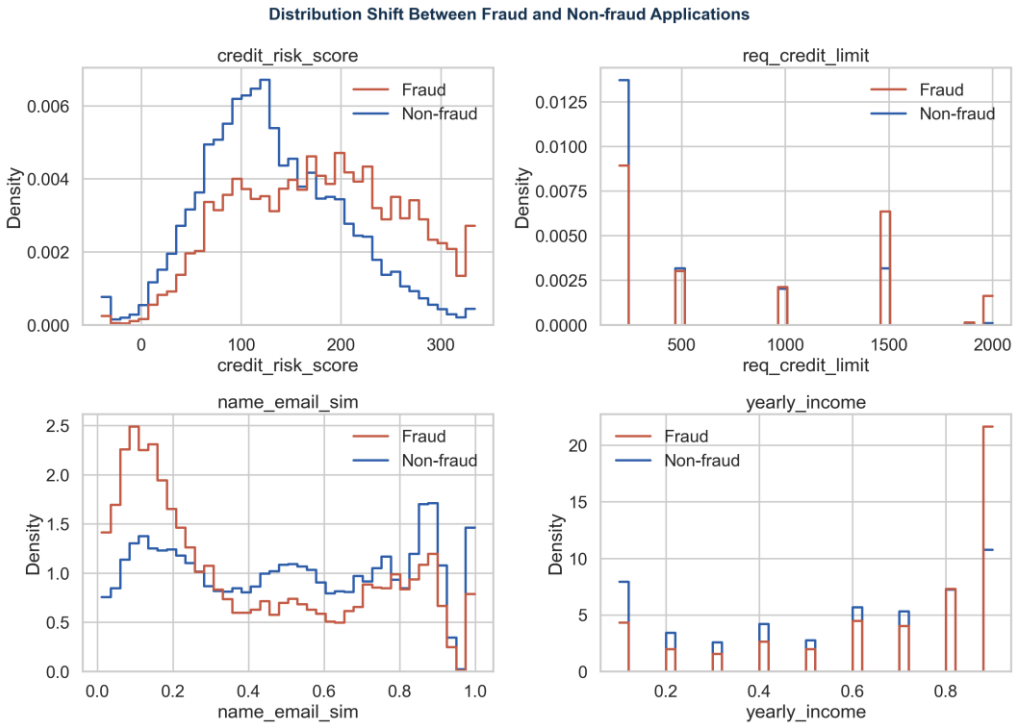
범주형 변수별로 **fraud rate 차이**가 뚜렷하게 나타났다.

- High-risk examples:
- payment_type = AC
- employment_status = CC
- housing_status = BA
- device_os = windows
- n= labels prevent over-interpreting tiny groups

Category identity itself carried meaningful fraud risk information.

Numeric Distribution Shift

Fraud and Non-Fraud Show Distribution Shift



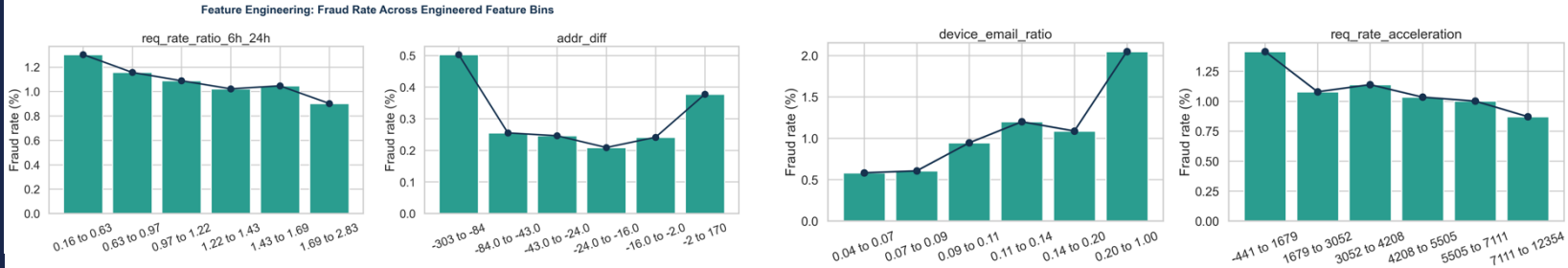
fraud와 non-fraud는 여러 **numeric feature**에서 다른 분포를 보였다.

- **Examples**:
- **credit_risk_score**: higher-risk region
- **req_credit_limit:** different usage profile
- **yearly_income**: shifted distribution
- **name_email_sim**: lower similarity in fraud

Fraud was not random noise; it showed consistent numeric distribution differences

Engineered Feature Validation

Engineered Features Captured Risk Patterns



직접 설계한 **engineered features**가 실제 fraud rate와 연결되었다.

- device_email_ratio ↑ → fraud risk ↑
- req_rate_ratio_6h_24h ↑ → short-term burst risk ↑
- req_rate_acceleration ↑ → abnormal request concentration ↑

Feature engineering was not cosmetic; it exposed interpretable fraud patterns.

Modeling Pipeline

Leakage-Aware Modeling Pipeline

전처리 단계:

-1 → NaN

missing indicator 추가

global feature engineering 적용

5-fold Stratified CV 기반 학습

fold의 train split에서만 생성한 통계형 feature:

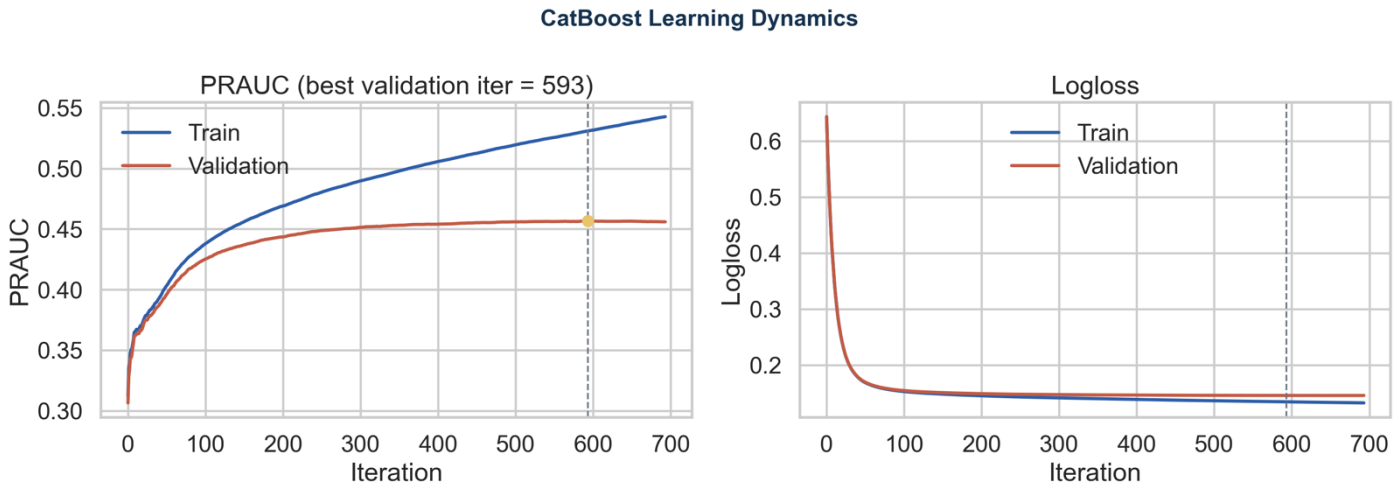
target encoding

group statistics

최종 예측은 **LightGBM + CatBoost ensemble**으로 생성

Training Behavior of CatBoost

CatBoost Training Behavior



Validation PRAUC는 **iteration 593**에서 최고점에 도달했다. (최고 validation PRAUC: **0.45665)**

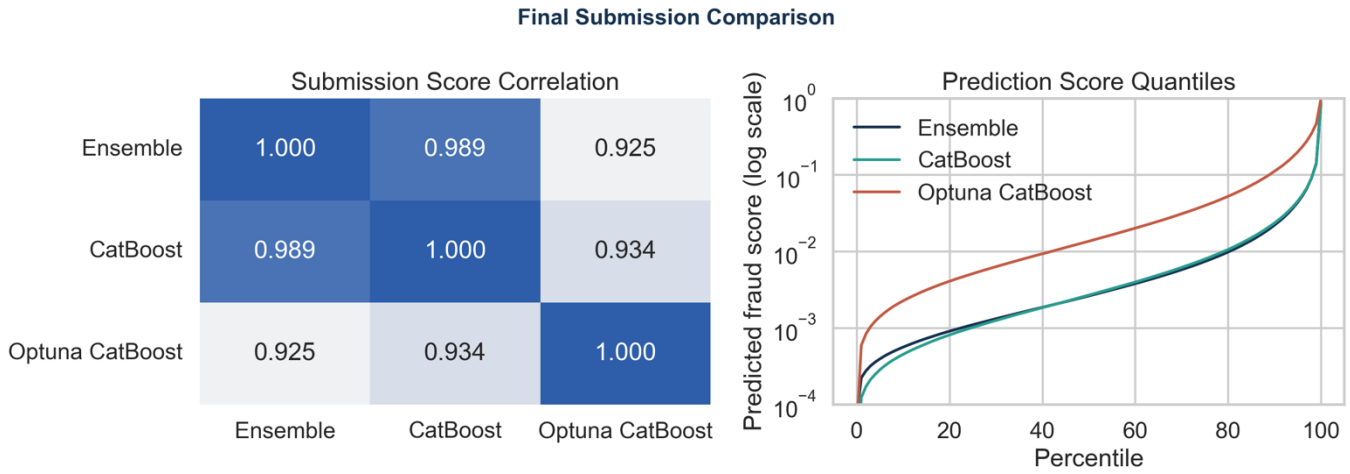
이후에는 train 성능만 더 좋아지고 validation은 거의 개선되지 않았다.

이는 충분한 학습 이후 **overfitting 구간**이 시작됨을 시사.

따라서 **early stopping**이 중요한 역할을 했다.

Submission Comparison

Why the Final Submission Was Chosen



- submission.csv and submission_cat.csv are highly correlated
- submission_optuna.csv is more aggressive in the upper score range
- Final choice considered:

  performance / stability / score distribution behavior

Conclusion

- train.csv 기반의 **fraud detection pipeline**을 구축하였다.
- 데이터 분석을 바탕으로 -1을 일부 column에서 **missing**으로 처리하였다.
- categorical feature와 engineered feature가 모두 강한 **risk signal**임을 확인하였다.
- **5-fold leakage-aware training**을 적용하였다.
- 최종 결과는 LightGBM + CatBoost ensemble submission.csv로 생성하였다.